Patterns in the Times Table

Look for patterns. They help you remember the facts.

The times table

×	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
4	4	8	12	16	20	24	28	32	36	40
5	5	10	15	20	25	30	35	40	45	50
6	6	12	18	24	30	36	42	48	54	60
7	7	14	21	28	35	42	49	56	63	70
8	8	16	24	32	40	48	56	64	72	80
9	9	18	27	36	45	54	63	72	81	90
10	10	20	30	40	50	60	70	80	90	100

Row 4 is all even:
4 × anything is even.

3 × 7 and 7 × 3 match:
turnarounds mirror across the table.

Going down a column:
add the top number each time.

Odd × odd is odd:
3 × 5 = 15, 7 × 9 = 63.

Double a number, double the answer

$3 \times 6 = 18$

$6 \times 6 = 36$

6 is double 3,
so 36 is double 18.

Your turn!

×4 4 8 **12** 16 **20** **24**

For teachers

Goal: find patterns in the times table and explain them (3.OA.9). Ask "Why is every number in the 4s row even?" and "What do you notice about 3 × 7 and 7 × 3?" Patterns make the hard facts easier to remember.

Patterns in the Times Table: Practice

Use the patterns. Check with the table if you need to.

Part 1 Fill in the row Add the same number each time.

1) ×3 3 **6** 9 12 **15** 18 **21** 24 27 **30**

2) ×6 6 12 **18** **24** 30 36 42 **48** **54** 60

3) ×8 8 **16** 24 32 **40** **48** 56 64 **72** 80

4) ×7 7 14 **21** 28 35 **42** **49** 56 63 **70**

5) ×9 9 **18** 27 **36** 45 54 63 **72** **81** 90

6) ×4 4 8 12 16 **20** 24 **28** **32** 36 **40**

Part 2 Pieces of the table Fill in the missing products.

7)

×	4	5	6	7
3	12	15	18	21
4	16	20	24	28
5	20	25	30	35

8)

×	2	3	4	5
6	12	18	24	30
7	14	21	28	35
8	16	24	32	40

Patterns in the Times Table: Practice (continued)

Part 2 Pieces of the table (continued)

9)

×	6	7	8	9
5	30	35	40	45
6	36	42	48	54
7	42	49	56	63

10)

×	5	6	7	8
7	35	42	49	56
8	40	48	56	64
9	45	54	63	72

Part 3 Even or odd? Even × any number is even. Odd × odd is odd.

11) 5×7 — even **(odd)**

12) 4×7 — **(even)** odd

13) 7×7 — even **(odd)**

14) 3×8 — **(even)** odd

15) 5×5 — even **(odd)**

16) 9×3 — even **(odd)**

17) 8×3 — **(even)** odd

18) 2×9 — **(even)** odd

19) 6×9 — **(even)** odd

Patterns in the Times Table: Practice (continued)

Part 4 Double it No help this time. Double one number, double the answer.

20) $2 \times 7 = 14$ so $4 \times 7 =$ **28**

21) $3 \times 9 = 27$ so $6 \times 9 =$ **54**

22) $4 \times 6 = 24$ so $8 \times 6 =$ **48**

23) $2 \times 8 = 16$ so $4 \times 8 =$ **32**

24) $3 \times 4 = 12$ so $6 \times 4 =$ **24**

25) $5 \times 7 = 35$ so $10 \times 7 =$ **70**

26) $4 \times 9 = 36$ so $8 \times 9 =$ **72**

27) $2 \times 6 = 12$ so $4 \times 6 =$ **24**

28) $3 \times 7 = 21$ so $6 \times 7 =$ **42**

29) $4 \times 8 = 32$ so $8 \times 8 =$ **64**

Patterns in the Times Table: Practice (continued)

Part 5 Times 9 from times 10 No help this time. Take away one group.

30) $10 \times 6 = 60$ so $9 \times 6 =$ **54**

31) $10 \times 3 = 30$ so $9 \times 3 =$ **27**

32) $10 \times 8 = 80$ so $9 \times 8 =$ **72**

33) $10 \times 4 = 40$ so $9 \times 4 =$ **36**

34) $10 \times 7 = 70$ so $9 \times 7 =$ **63**

35) $10 \times 9 = 90$ so $9 \times 9 =$ **81**

Part 6 Times 5 from times 10 No help this time. Times 5 is half of times 10.

36) $10 \times 8 = 80$ so $5 \times 8 =$ **40**

37) $10 \times 3 = 30$ so $5 \times 3 =$ **15**

38) $10 \times 6 = 60$ so $5 \times 6 =$ **30**

39) $10 \times 9 = 90$ so $5 \times 9 =$ **45**

40) $10 \times 4 = 40$ so $5 \times 4 =$ **20**

41) $10 \times 7 = 70$ so $5 \times 7 =$ **35**

Patterns in the Times Table: Practice (continued)

Part 7 Double it

42) $2 \times 2 = 4$ so $4 \times 2 =$ **8**

43) $3 \times 6 = 18$ so $6 \times 6 =$ **36**

44) $4 \times 2 = 8$ so $8 \times 2 =$ **16**

45) $3 \times 5 = 15$ so $6 \times 5 =$ **30**

46) $5 \times 2 = 10$ so $10 \times 2 =$ **20**

47) $3 \times 2 = 6$ so $6 \times 2 =$ **12**

48) $5 \times 3 = 15$ so $10 \times 3 =$ **30**

49) $4 \times 7 = 28$ so $8 \times 7 =$ **56**

50) $2 \times 9 = 18$ so $4 \times 9 =$ **36**

51) $4 \times 3 = 12$ so $8 \times 3 =$ **24**

52) $3 \times 3 = 9$ so $6 \times 3 =$ **18**

53) $5 \times 8 = 40$ so $10 \times 8 =$ **80**

Patterns in the Times Table: Practice (continued)

Part 7 Double it (continued)

54) $2 \times 5 = 10$ so $4 \times 5 =$ **20**

55) $5 \times 4 = 20$ so $10 \times 4 =$ **40**

56) $3 \times 8 = 24$ so $6 \times 8 =$ **48**

57) $4 \times 5 = 20$ so $8 \times 5 =$ **40**

58) $5 \times 5 = 25$ so $10 \times 5 =$ **50**

59) $2 \times 3 = 6$ so $4 \times 3 =$ **12**

60) $5 \times 6 = 30$ so $10 \times 6 =$ **60**

Part 8 Even or odd?

61) 4×9

(even) odd

62) 6×4

(even) odd

63) 6×7

(even) odd

Patterns in the Times Table: Practice (continued)

Part 8 Even or odd? (continued)

64) 2×7

(even) odd

65) 7×6

(even) odd

66) 3×2

(even) odd

67) 3×3

even (odd)

68) 7×2

(even) odd

69) 6×6

(even) odd

70) 2×3

(even) odd

71) 4×2

(even) odd

72) 2×2

(even) odd

73) 6×5

(even) odd

74) 2×8

(even) odd

75) 4×3

(even) odd

Patterns in the Times Table: Practice (continued)

Part 8 Even or odd? (continued)

76) 7×8 (even) odd

77) 5×2 (even) odd

78) 6×8 (even) odd

Part 9 Remember? Problems from earlier packets.

79) 7 m by 6 m: area = 42 sq m

80) 40 minutes after 2:35 = 3:15

81) Round 481 to the nearest hundred: 500

82) $3 \times 70 = 210$

83) $72 - 38 = 34$

84) $64 - 27 = 37$

Part 10 Word problems Look for a pattern.

85) Each table has 4 legs. Is the number of legs on 7 tables even or odd?

Number sentence 7 × 4 = 28, even

Answer even

Draw or show your work

86) A pack has 6 juice boxes. 3 packs have 18. How many juice boxes are in 6 packs?

Number sentence double 18 = 36

Answer 36 juice boxes

Draw or show your work

Patterns in the Times Table: Practice (continued)

Part 10 Word problems (continued)

87) 9 boxes hold 8 crayons each. How many crayons is that?

Number sentence **10 × 8 = 80, 80 − 8 = 72**

Answer **72** crayons

Draw or show your work

88) Ben counts by 5s, 8 times. What number does he say last?

Number sentence **8 × 5 = 40**

Answer **40**

Draw or show your work

89) Challenge: Kim says 7 × 9 = 64. Odd × odd is odd, so that can't be right. What is 7 × 9?

Number sentence **7 × 9 = 63, odd**

Answer **63**

Draw or show your work

Check-In: Patterns in the Times Table

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 Fill in the row

1) ×4 4 8 **12** **16** 20 24 28 **32** 36 **40**

Part 2 Pieces of the table

2)

×	5	6	7	8
6	30	36	42	48
7	35	42	49	56
8	40	48	56	64

3)

×	3	4	5	6
6	18	24	30	36
7	21	28	35	42
8	24	32	40	48

Part 3 Even or odd?

4) 8×9 (even) odd

5) 7×5 even (odd)

6) 5×8 (even) odd

Part 4 Double it

7) $4 \times 4 = 16$ so $8 \times 4 =$ **32**

8) $2 \times 4 = 8$ so $4 \times 4 =$ **16**